WORKED SOLUTION

STEP III 2011, Question 4

Pure • Young's Inequality

STEP III 2011, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 3.4, *Areas That Prove Inequalities*: https://beyondstep.keithmcnulty.org/ch03/areas-inequalities.html

The key idea

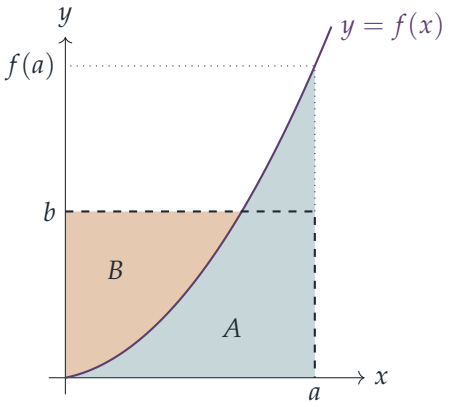
Both integrals in $(*)$ are areas on the same picture: $\int_0^a f(x)\,\mathrm{d}x$ is the area between the curve and the x-axis, and $\int_0^b f^{-1}(y)\,\mathrm{d}y$ is the area between the curve and the y-axis. Together they always cover the rectangle with corners $(0,0)$ and (a,b), with some to spare unless the corner (a,b) sits on the curve. Parts (ii) and (iii) are then just a matter of choosing f and doing two integrals.

Solution

Part (i)

Since f is continuous with positive gradient, it is strictly increasing, so f^{-1} exists and is also increasing, and $f(0) = 0$ means the curve $y = f(x)$ starts at the origin.

Write A for the region between the curve and the x-axis for $0 \leq x \leq a$; its area is $\int_0^a f(x)\,\mathrm{d}x$. Write B for the region between the curve and the y-axis for $0 \leq y \leq b$; integrating with respect to y, its area is $\int_0^b f^{-1}(y)\,\mathrm{d}y$. The two regions lie on opposite sides of the curve, so they don't overlap.



Suppose first that $b \le f(a)$, as in the picture. Every point of the rectangle $0 \le x \le a$, $0 \le y \le b$ lies either on or below the curve (so in A) or on or to the left of it (so in B). So A and B together cover the rectangle, of area ab, and $(*)$ holds. The surplus is the part of A above the line $y = b$, the sliver between the line and the curve for $f^{-1}(b) \le x \le a$.

If instead $b \ge f(a)$, the same argument works with the roles swapped: the rectangle is covered, and now the surplus is the part of B to the right of the line $x = a$.

So $(*)$ holds in every case, and equality needs the surplus to have zero area. When $a, b > 0$ and $b \ne f(a)$, the surplus is bounded by the curve and a straight line that meet at only one end, so it has positive area. Hence, for $a > 0$ and $b > 0$,

$$\text{equality holds if and only if } b = f(a),$$

that is, when the corner (a, b) of the rectangle lies on the curve.

Part (ii)

For $p > 1$, $f(x) = x^{p-1}$ is continuous on $x \ge 0$, has $f(0) = 0$, and is strictly increasing there, so $(*)$ applies. (Its gradient $(p-1)x^{p-2}$ is positive for $x > 0$; if $p < 2$ it blows up at 0, which doesn't affect the area argument.) Its inverse is $f^{-1}(y) = y^{1/(p-1)}$.

The condition $\frac{1}{p} + \frac{1}{q} = 1$ gives $\frac{1}{q} = \frac{p-1}{p}$, so

$$q = \frac{p}{p-1}, \qquad \frac{1}{p-1} + 1 = \frac{p}{p-1} = q.$$

So $f^{-1}(y) = y^{q-1}$, and the two integrals are

$$\int_0^a x^{p-1}\,\mathrm{d}x = \frac{a^p}{p}, \qquad \int_0^b y^{q-1}\,\mathrm{d}y = \frac{b^q}{q}.$$

Substituting into $(*)$ gives

$$ab \le \frac{a^p}{p} + \frac{b^q}{q},$$

as required.

Ask yourself What does the equality condition $b = f(a)$ say here, and how do we simplify b^q?

The condition is $b = a^{p-1}$. Then $b^q = a^{(p-1)q} = a^p$, because $(p-1)q = p$. So

$$\frac{a^p}{p} + \frac{b^q}{q} = a^p\left(\frac{1}{p} + \frac{1}{q}\right) = a^p = a \cdot a^{p-1} = ab,$$

and equality does hold.

Part (iii)

Take $f(x) = \sin x$ on $0 \le x \le \frac{1}{2}\pi$. It is continuous, $f(0) = 0$, and its gradient $\cos x$ is positive for $0 \le x < \frac{1}{2}\pi$, so it is strictly increasing (the gradient vanishes only at the endpoint, which does not affect the argument). It maps $[0, \frac{1}{2}\pi]$ onto $[0, 1]$, with inverse arcsin, and these are exactly the ranges allowed for a and b.

The first integral is

$$\int_0^a \sin x \, \mathrm{d}x = 1 - \cos a.$$

For the second, integrate by parts with $u = \arcsin y$ and $\mathrm{d}v = \mathrm{d}y$:

$$\begin{aligned}\int_0^b \arcsin y \, \mathrm{d}y &= \Big[y \arcsin y\Big]_0^b - \int_0^b \frac{y}{\sqrt{1-y^2}} \, \mathrm{d}y \\ &= b \arcsin b + \Big[\sqrt{1-y^2}\Big]_0^b \\ &= b \arcsin b + \sqrt{1-b^2} - 1.\end{aligned}$$

Adding, the 1s cancel and $(*)$ becomes

$$ab \le b \arcsin b + \sqrt{1-b^2} - \cos a,$$

which is the required inequality, with equality when $b = \sin a$.

Ask yourself We have a free choice of a. Which one gives the inequality we want?

For $t \ge 1$, put $b = t^{-1}$, which lies in $(0, 1]$ as needed. Multiplying the inequality by $t > 0$ and using $t\sqrt{1-t^{-2}} = \sqrt{t^2-1}$ (valid because $t > 0$) gives

$$a \le \arcsin(t^{-1}) + \sqrt{t^2-1} - t\cos a \quad \text{for every } a \in [0, \tfrac{1}{2}\pi].$$

The target has no a in it and has a t on the right, which is what $t\cos a$ becomes at $a = 0$. Taking $a = 0$:

$$0 \le \arcsin(t^{-1}) + \sqrt{t^2-1} - t, \qquad \text{that is,} \qquad \arcsin(t^{-1}) \ge t - \sqrt{t^2-1} \quad (t \ge 1).$$

Checks

- At $t = 1$ the final inequality reads $\frac{1}{2}\pi \geq 1$, and at $t = 2$ it reads $\frac{1}{6}\pi \approx 0.5236 \geq 2 - \sqrt{3} \approx 0.2679$.
- As $t \to \infty$, both sides are of order $\frac{1}{t}$: $\arcsin(t^{-1}) \approx t^{-1} + \frac{1}{6}t^{-3}$, while $t - \sqrt{t^2-1} = \frac{1}{t+\sqrt{t^2-1}} \approx \frac{1}{2t}$. So the bound is about half the true value.
- We tested the part (iii) inequality at 2×10^5 random points (a, b); it never failed, and the gap shrank to zero only near the curve $b = \sin a$. We also checked Young's inequality for random p, a and b, and checked both integrals by Simpson's rule.

Common slips

- Only drawing the case $b \leq f(a)$ in part (i). The case $b \geq f(a)$ is symmetric, but you should say so.
- Writing $f^{-1}(y) = y^{1/(p-1)}$ and then not spotting that $1/(p-1) = q - 1$, so the integral b^q/q doesn't appear.
- In the last step, looking for the a that optimises the bound. That a is $\arcsin(t^{-1})$, which only gives the trivial identity. The simplest choice, $a = 0$, is the one that works.

Beyond STEP

Part (ii) is Young's inequality, the standard route to Hölder's inequality $\sum |x_i y_i| \leq \left(\sum |x_i|^p\right)^{1/p} \left(\sum |y_i|^q\right)^{1/q}$, which in turn underlies the L^p spaces of analysis. The pairing of $\int f$ with $\int f^{-1}$ is the germ of the Legendre transform and convex duality, which reappear in mechanics (moving from the Lagrangian to the Hamiltonian) and in optimisation. For the analysis background, see Bartle and Sherbert, *Introduction to Real Analysis*.

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